



Blowing in the late Cenozoic wind—detrital zircon river contributions to an interior Alaska loess deposit

Elisabeth Nadin ^{a,*}, Scott Goddard ^a, Jeff Benowitz ¹, Paul O'Sullivan ^b

^a University of Alaska Fairbanks, USA

^b GeoSep Services, USA



ARTICLE INFO

Article history:

Received 2 July 2021

Received in revised form

19 October 2021

Accepted 29 October 2021

Available online 30 November 2021

Handling Editor: C. O'Cofaigh

Keywords:

Detrital zircon

Loess

Interior Alaska

Winds

River zircon ages

ABSTRACT

The extensive Interior Alaska loess deposits record glacial advance and retreat over the past 3 Myrs. Refining our understanding of the still-debated origins of this loess will help us refine our understanding of climate patterns since the late Cenozoic. We present here the first detrital-zircon ages from the Gold Hill loess, a well-studied deposit in Fairbanks, Alaska, as well as new detrital-zircon ages from local fluvial sediments. We describe a new statistical approach to quantify the contributions of sediment loads from local rivers to account for the grain-age distributions within the Fairbanks loess. We find that zircons from the 3 Ma base of the loess deposit come mainly from sediments carried by the Tanana and Delta rivers, and Yukon River sediments contributed more zircons over time as this river system developed. This indicates that loess-depositing winds in Interior Alaska came from the east/southeast, a pattern that persists today. Although we cannot exclude other sources, our new statistical approach suggests that a mix of sediments carried from local river deposits can account for almost all of the zircon grains in the Gold Hill loess deposit of interior Alaska.

© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

As we look toward a future that may include an ice-free Arctic Ocean that would influence wind patterns, it becomes increasingly important to understand wind directions of an ice-free past. Interior Alaska is home to one of North America's most extensive deposits of loess, an eolian sediment, that preserves several episodes of deposition marking glacial advance and retreat since late Cenozoic time. The precise origins of this loess are still debated, and the relatively continuous ca. 3 Ma–Holocene deposits of Interior Alaska (Westgate et al., 1990; Begét, 1996; Preece et al., 1999; Berger, 2003) have been studied to reconstruct climate variations, particularly wind patterns, of the interglacial and glacial periods it records. Local Tanana and Yukon river deposits (Fig. 1) have been proposed as the sources from which the loess was derived Muhs et al. (2003).

This study follows others aimed at quantifying wind directions in Interior Alaska (Table 1). These include studies of (1) magnetic mineral fabric suggesting exclusively northerly paleowinds (Lagroix and Banerjee, 2002); (2) thinning direction of loess

deposits indicating exclusively southerly paleowinds (Péwé, 1955); (3) northeasterly winds for the 25–12 kya interval, inferred from dune orientations (Hopkins et al., 1982; Lea and Waythomas, 1990); (4) Holocene southerly winds indicated by sand sheet location (Froese et al., 2005); and (5) last-glacial katabatic winds from the Alaska Range to the south (Thorson and Bender, 1985; Guthrie, 1990; Muhs et al., 2018).

Fine airborne particulates can travel far in Alaska—volcanic ash from beds throughout Interior Alaska loess deposits have been traced to the Aleutian arc, eastern Alaska Peninsula, and Wrangell volcanic field (Preece et al., 1999), suggesting wind-blown sediment contributions from as far away as 1000 km. Despite this, most studies of the loess support more local sources. Muhs and Budahn (2006) suggest that in late Quaternary time, rivers draining the Alaska Range, Yukon-Tanana Uplands, and Brooks Range (Fig. 1) all contributed to the central Alaskan loess deposit, and wind patterns varied during slow accumulations of loess. We pursue this interpretation of loess deposited by prevailing winds that entrained and distributed glacial dust delivered locally by rivers that drain bedrock throughout Alaska (Péwé, 1955, 1975; Begét, 1996; Muhs and Budahn, 2006; Muhs et al., 2018). We present here the first detrital-grain study of zircon from interior Alaska loess, alongside detrital-zircon age distributions from local riverbeds, in order to

* Corresponding author.

E-mail address: enadin@alaska.edu (E. Nadin).

¹ Independent Scientist



Fig. 1. Map of Alaska showing major river systems and mountainous regions, as well as the loess sampling site (Gold Hill) and the four river sample sites (Yukon, lower Tanana, Delta, and Nenana). The sampled rivers are shown in different colors to distinguish them from unsampled tributaries (shown in black), and matched to sample line colors in figures 4 and 7. The Yukon–Tanana Uplands is shaded in light green.

test previous hypotheses and construct a paleo-depositional scenario.

Detrital zircons are commonly used to track the provenances and provide age brackets for sedimentary deposits, and are particularly useful in resolving mixed sources. Recently, detrital-zircon analyses have been widely utilized in loess provenance studies in China (Stevens et al., 2010; Pullen et al., 2011; Xie et al., 2012; Che and Li, 2013; Liu et al., 2014; Licht et al., 2016; Gong et al., 2017; Shang et al., 2018; Sun et al., 2018; He et al., 2020; Wu et al., 2021), and more rarely in Europe (Újvári et al., 2012; Nawrocki et al., 2018). Limited detrital-grain geochronologic studies of U.S. loess deposits include the Peoria Loess (Aleinikoff et al., 1999, 2008) and two loessites from Utah and New Mexico (Soreghan et al., 2012). This study is the first to use detrital-zircon age distributions to examine the origins of the significant loess deposit near Fairbanks, Alaska.

In addition to determining grain-age populations that may reflect source components for this important record of North American glacial cycles, we present a new statistical approach to quantifying source contributions to detrital-grain populations. In recent years, statistical analysis has emerged as a more robust means of assessing similarities among larger and more diverse grain-age populations (e.g., Gehrels, 2012; Vermeesch, 2013, 2018; Saylor and Sundell, 2016; Sundell and Saylor, 2017; Saylor et al., 2019; Tye et al., 2019). Many of these rely on constructing probability density plots (PDPs), which we also do here. We model the cumulative distribution function (CDF) of detrital-zircon populations from four loess intervals using a linear combination of the possible parent (river) sources. We thereby estimate contributions

to the loess over time from the local Tanana, Delta, Nenana, and Yukon river systems.

1.1. Gold Hill loess, Fairbanks, Alaska, USA

Loess deposits in the central latitudes of the U.S. are generally associated with glacial–interglacial cycles (e.g., Aleinikoff et al., 1999). In central Alaska, the continuous presence of glaciers throughout the Quaternary Period in the neighboring Alaska Range and other mountains led to a continuous supply of glacial silt to the ice-free interior. Currently, loess deposition in Interior Alaska takes place during the snow-free spring to fall months. There is a particularly dusty period caused by katabatic winds in late May to June in Fairbanks (Fig. 2; see also Discussion section). However, modern-day atmospheric dust measurements indicate a semi-continuous flux (Bégét, 1996). Alaskan loess therefore is thought to provide a valuable semi-continuous record of both glacial and interglacial cycles of the Quaternary Period.

The Gold Hill loess is the oldest, thickest, and most extensive loess deposit in Interior Alaska (cf. Péwé et al., 2009 detailed study on this deposit). A number of stratigraphic sections of Gold Hill loess are described by Preece et al. (1999) and Péwé et al. (2009). We sampled along a 40-m-tall exposure west of Fairbanks near the town of Ester, known as the Gold Hill Cut (Fig. 3; Westgate et al., 1990; Preece et al., 1999; Péwé et al., 2009). This exposure has been fairly well described because of how relatively undisturbed and thick it is, and because the deposit in this location preserves fossils, unconformities marking major environmental changes, and tephra that have been dated (Preece et al., 1999; Péwé et al., 2009).

The lowest ash layer in the Gold Hill section, the PA tephra, is cited as 2.02 Ma in Péwé et al. (2009, after Preece et al., 1999). This tephra was also dated by Westgate et al. (1990) to be 1.89 Ma. For the purposes of our study, we call this ash bed ca. 2 Ma. Additional timing constraints, particularly stratigraphic, paleomagnetic, thermoluminescence, and glass-shard fission-track studies, have placed the base of the Gold Hill cut exposure at ca. 3 Ma (Westgate et al., 1990; Preece et al., 1999), and the top at younger than 30 ka (Berger and Péwé, 2001; Berger, 2003; Davies et al., 2016; Jensen et al., 2016). From the bottom of the section upward (and marked on Fig. 3), dated ash layers include the PA tephra (ca. 2 Ma) and the SP tephra (dated to ca. 0.9 Ma by Westgate et al., 1990 and Preece et al., 1999, and referred to in this study as ca. 1 Ma). Detailed (5-cm-interval) sampling of this section revealed a well-preserved horizon of the 0.9 Ma Kamikatsura event—an established magnetostratigraphic marker—in the middle of the exposure; its undisturbed nature supports the interpretation that the loess sequence is intact here (Evans et al., 2011). The top of the Gold Hill section is well dated because it preserves strata of environmental significance in its top 12 m (cf. Berger, 2003). This interval is reportedly undisturbed (Péwé unpublished notes, 1987, cited by Berger, 2003). Three meters below the top of the section, an organic-rich soil level is thought to mark a global thermal event that may have even thawed local permafrost (cf. Péwé et al., 2009). Thermoluminescence ages from just above this soil level place its development ca. 30 ka (Berger and Péwé, 2001; Berger, 2003). This is supported by the presence of the Dawson tephra, which is located at a similar stratigraphic level as the soil (Jensen et al., 2016) and was dated to ca. 30 ka (Froese et al., 2006; Preece et al., 2011; Davies et al., 2016). Additionally, a magnetic susceptibility high in this location likely corresponds with the same signal interpreted to represent the last glacial maximum elsewhere in central Alaska, dated to ca. 16 ka (Dortch et al., 2010; Jensen et al., 2016). Since we sampled below the very top of the deposit, we can constrain our youngest loess sample to less than 16 ka, and refer to it as Holocene.

With the aim of evaluating differences in loess sources at four

Table 1

Summary of past wind history and sediment source interpretations for the Gold hill loess deposit.

Time interval	Wind Direction/Strength	Loess source	Method	Reference
250–0 ka	Low strength during interglacial periods; higher during glacial periods.	N/A	magnetic susceptibility variations	Begét and Hawkins, 1989
150–0 ka	NW–SE during glacial periods; N–S during interglacials.	N/A	anisotropy of magnetic susceptibility (AMS)	Lagroix & Banerjee (2002)
modern; <160 ka	NNE–SSW summer = past interglacials; E–W winter = past glacials.	N/A	climatological reanalysis; AMS	Lagroix and Banerjee (2004a), b
Holocene	Strongest winds from S during deposition; coincident with modern winds.	fluvial deposits of Yukon, Nenana, and Tanana rivers; greatest contribution from Tanana River	ratios of mobile to immobile elements	Muhs & Budahn (2006)
Holocene	Strongest winds from N.	N/A	sand-dune orientation	Collins (1985); Lea (1996)
19–12 ka	Strongest winds from N; katabatic winds from S during glacier expansion.	Yukon–Tanana Upland loess mostly sourced from Yukon R; Tanana R. valley was distal source developed during glacier expansion	particle-size analysis; bulk mineralogy; major and trace-element ratios	Muhs et al. (2018)
last glacial period	Strong winter winds from E/NE; weaker summer winds from S.	N/A	climate modeling	Alder & Hostetler (2015)
3 Ma–Holocene	Winds predominantly from S/SE, with contribution from NE.	3 Ma: 45% Tanana–Delta; 37% Yukon; 18% Nenana. 2 Ma: 72% Tanana–Delta; 23% Yukon; 5% Nenana. 1 Ma: 45% Tanana–Delta; 40% Yukon; 15% Nenana. Holo: 40% Tanana–Delta; 38% Yukon; 22% Nenana	Detrital zircon U–Pb ages	This study

**Fig. 2.** Photograph of windborne dust over the Tanana River valley in Fairbanks, looking south toward the high peaks of the Alaska Range. Photo taken by W. Tape on May 11, 2012.

different, well-constrained, time intervals, we sampled near the ca. 3 Ma base, just above the ca. 2 Ma PA tephra, just above the ca. 1 Ma SP tephra, and just below the top of the Gold Hill Cut (Fig. 3). We were able to access the ca. 3 Ma base through a shaft dug by a periglacial processes class at University of Alaska Fairbanks.

1.2. Former interpretations of wind histories

As described below and summarized in Table 1, several studies support the idea that the vast distribution and thicknesses of Interior Alaska loess indicate varying wind strengths and directions throughout the Pleistocene period. Multiple wind directions have been interpreted from loess and surrounding features. Lagroix and Banerjee (2002, 2004b) suggested that long-term prevailing wind direction in central Alaska changed from northwest–southeast

during glacial periods to north–south during interglacials of the past 150 ka. Based on their calculations of wind vectors from 1968 to 1996, Lagroix and Banerjee (2004a, b) further posit that modern north-northeast–south-southwest summer wind directions represent past interglacials, while modern east–west winter wind directions represent those prevalent in glacial periods.

Muhs and Budahn (2006) suggested three sources of Interior Alaska loess during the last glacial period (see Fig. 1 for locations): (1) fluvial deposits of the Yukon River, which drains the southern flank of the eastern Brooks Range, interior Yukon, and the northern flank of the Yukon–Tanana Upland; (2) fluvial deposits of the Nenana River, which drains a limited portion of the northern side of the Alaska Range; and (3) fluvial deposits of the Tanana River, which drains the southern flank of the Yukon–Tanana Upland, the Wrangell Mountains, and a portion of the northern side of the Alaska Range. (In this study, we test detrital-zircon ages from these three suggested sources, as well as the Delta River.) Muhs and Budahn (2006) interpret a strong Tanana-River contribution to the loess, and place prevailing winds from the south during loess deposition, coincident with modern summertime winds. This interpretation contrasts with north-sourced winds during loess transport that were interpreted from sand-dune orientations (Collins, 1985; Lea, 1996). Muhs et al. (2018) refined their findings to suggest that Yukon–Tanana Upland loess deposited between 19 and 12 ka was mostly sourced from the Yukon River to the north, resulting from regional winds. The Tanana River valley was a secondary source to the south. This contrast in wind direction and strength is supported by climate modeling of the last glacial period—Alder and Hostetler (2015) show that winter surface paleowinds were stronger (and much stronger than present) and from the east/northeast, while summer winds were weaker (though still stronger than today's) and from the south.

The studies described above reconstitute wind directions through the last glacial age. In this work, we calculate river contributions to Interior Alaska loess dating back to ca. 3 Ma, into the Pliocene epoch. We support previous suggestions that the loess was built from wind-reworked river sediments, giving insight into how these contributions shifted over time.

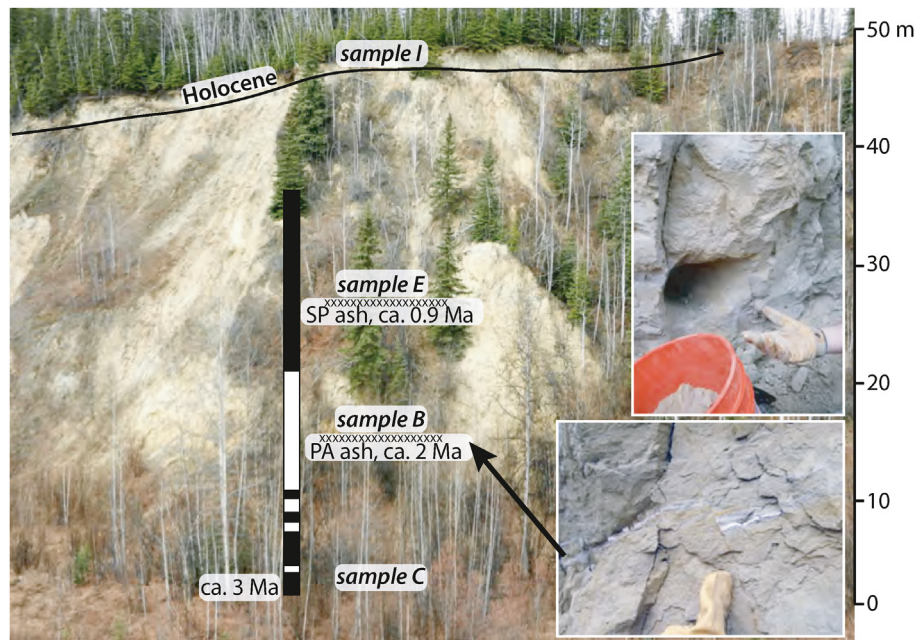


Fig. 3. Outcrop photo of the Gold Hill loess site, showing sample sites (from bottom: C, B, E, I), tephrochronology (Westgate et al., 1990; Preece et al., 1999; Péwé et al., 2009), and the magnetostratigraphic age constraints shown in Péwé et al. (2009) Fig. 6 (black and white bar, scaled directly to the height and ash intervals of our sampling site). The bottom inset photo shows the bright PA ash layer, and the top inset photo shows our sampling method of digging into the loess, just above the dated ash layer, to avoid surface contamination.

1.3. Detrital grains as source indicators

Detrital zircon U–Pb ages are commonly used to determine the bedrock provenance of sedimentary deposits. Generally, sample-age peaks from ~100 dated grains are matched to distinct ages of geologically plausible bedrock sources. Over the past several years, the statistical robustness of this approach has been tested, with notable studies indicating (1) a sample size of 117 dated grains yields 95% confidence that no age peaks that constitute a share of >0.05 of the sampled population will be missed (Vermeesch, 2004); and (2) there is little difference in identified major age peaks for samples with $n = 60, 100$, or 300 grains (Tye et al., 2019). We use detrital zircon U–Pb ages to evaluate the Yukon, Tanana, Delta, and Nenana river (Fig. 1) sediment as potential sources of the Gold Hill loess. Due to its apparently eolian transport mechanism, the Gold Hill loess is likely to be derived from glacio-fluvial sediment rather than directly from bedrock. Our goal is to test if the grain-age populations of local rivers can account for the grain-age populations in the loess.

Loess deposits around the globe are attributed to water and wind transport, rather than direct glacial transport. It is a long-held view that the loess was brought to Interior Alaska by long-term eolian deposition, with minimal reworking, and particularly from the lower portion of dust clouds that redistributed local stream sediments (Begét, 1988), possibly up to hundreds of kilometers (Begét, 2001). Some loess deposits have been traced to the Tanana River floodplain (e.g., Begét et al., 1990; Muhs and Budahn, 2006). Local redistribution has been indicated for various other loess deposits as well, for example in the Colorado Plateau where detrital-zircon ages of a loessite were well-summarized as reflecting “the influence of transcontinental fluvial repositioning of derivative sediment” (Dickinson and Gehrels, 2003). Indeed, a Chinese Loess Plateau zircon study advises against matching loess detrital zircons to bedrock sources—up to 70 percent of the loess zircon grains came from wind-reworked Yellow River sediments (Licht et al., 2016). Nawrocki et al. (2018) concluded that the detrital-zircon

grains in the loess they studied were transported from the bedrock source by a local river before being wind transported to the loess deposit.

2. Methods

We collected four loess samples from the 50-m-tall Gold Hill Cut (Fig. 3) near the Trench 3 site just west of Fairbanks, which is described in Péwé et al. (2009). We sampled the base, just above each of two dated ash beds, and the top (Table 2). At each sampled interval, we filled half of a five-gallon bucket with silt after scraping off the exposed surface and digging several inches into the cliff face in order to avoid surface contamination. We also filled buckets with sediment at each of four interior Alaska rivers: the Tanana, Delta, Nenana, and Yukon (Fig. 1).

For all samples, we covered the sediment in the buckets with water and swirled and decanted suspended silt. We sent samples of the settled sandy silt to GeoSep Services (GSS) for zircon separation and analysis. Between 84 and 109 individual zircons from each sample were dated by the LA-ICP-MS method. Mineral grains were isolated and prepared for dating using standard procedures combined with specific customized procedures described by Donelick et al. (2005). See Supplement 1 for a full description of procedures to produce and process the zircon U–Pb data. Our ages are reported along with absolute errors of $\pm 2\sigma$ in Supplementary Data Table 1.

According to Vermeesch (2004), our sample sizes indicate that there is a 95% certainty that our samples may miss a fraction of 0.06–0.07 of the populations we tested.

2.1. Data analysis

Data analysis began with a qualitative comparison of normalized and cumulative age probability plots of each individual sample, created using the Excel-based tools provided by G. Gehrels (Arizona LaserChron Center). We computed the statistical

Table 2

Sample numbers and descriptions of intervals in the Gold Hill loess sampling site. Sampled intervals are in bold.

Sample	Description	height from base
110608-A	"PA" ash, ca. 2 Ma	~15 m
110608-B	Loess sample from just above PA	just above PA
110608-C	Bottom of exposed section, ca. 3 Ma	~1m
110608-D	Loess fractured and rooted, maybe slumped	~25 m
110608-E	Loess sample from near "SP" ash	~26m
110608-F	Loess sample from near HP ash	~30 m
110608-G	"HP" ash	~30 m
110608-H	perlicious cliff face	~31 m
110608-I	Loess sample from near top of section	~50m
Top of Section		~51 m

Bold indicates sample detrital ages reported in this study.

similarities amongst the loess samples and amongst the river samples using the Anderson-Darling test (cf. Vermeesch, 2018). *P*-values below 0.05 indicate that samples can be confidently distinguished from each other (Table 3).

Having tried to match various combinations of bulk river grains (for example, combining Delta and Tanana river samples) to a single loess interval age spectrum (Fig. 5), we recognized the value of a model that could unmix detrital age distributions. Our goal was to determine viable percentages of zircon contributions from each particular river "source". Initially we closely followed the emerging literature on statistical analysis of detrital zircon ages, particularly Saylor and Sundell (2016) and the subsequent Sundell and Saylor (2017) DZMix model. These efforts involved the computation of PDPs which smoothe sampled grain ages over their estimated 2 σ measurement errors.

At length, however, we postulated that a probabilistic analysis could possess two unique advantages over the trial-and-error approach of the DZMix method. Firstly, it might find better estimated mixture weights utilizing the additional information contained in the probability models invoked. Secondly, it would give

standard errors for those estimated mixture weights.

Therefore, we opted to devise a new probabilistic method that could optimize the match between a cumulative distribution function (CDF) of ages from a loess interval and a weighted combination of river grain ages. We began by creating probability density plots (PDPs) from age data for each interval that incorporates geochronological age uncertainties, as described in Saylor and Sundell (2016). The PDPs were computed at each point of a regular grid of length 100 stretched across the span of 0–3000 Ma (Fig. 4b, d).

In our new statistical approach, described below, we numerically integrated the PDPs to result in what one might think of as "smoothed" (as opposed to "empirical") cumulative distribution functions (hereafter simply "CDFs," Figs. 4 and 5), and opted to fit a probability model to these CDFs in place of their PDPs due to their monotonicity and higher degree of functional smoothness. We also created a multi-dimensional scaling (MDS) representation of how closely each sample set was related to each other (Fig. 6). This was based on the methodology applied to detrital-zircon chronology by Vermeesch (2013).

Table 3Results of the Anderson–Darling statistical test for similarities amongst detrital-zircon ages of the four river sediment samples and four loess intervals. *P*-values above 0.05 indicate insignificant evidence that the samples were drawn from distinct sources.

Anderson-Darling test statistics for rivers				
	Yukon	Nenana	Delta	Tanana
Yukon	0	6.36	6.64	2.98
Nenana	6.36	0	5.51	2.82
Delta	6.64	5.51	0	2.07
Tanana	2.98	2.82	2.07	0
Anderson-Darling <i>p</i> -values for rivers				
	Yukon	Nenana	Delta	Tanana
Yukon	1	0.00	0.00	0.03
Nenana	0.00	1	0.00	0.03
Delta	0.00	0.00	1	0.08
Tanana	0.03	0.03	0.08	1
Anderson-Darling test statistics for loess samples				
	B (2 Ma)	C (3 Ma)	E (0.9 Ma)	I (Holocene)
B	0	1.05	1.88	2.03
C	1.05	0	0.69	0.44
E	1.88	0.69	0	0.72
I	2.03	0.44	0.72	0
Anderson-Darling <i>p</i> -values for loess samples				
	B	C	E	I
B	1	0.33	0.11	0.09
C	0.33	1	0.57	0.81
E	0.11	0.57	1	0.54
I	0.09	0.81	0.54	1

2.2. Bayesian linear model

Each loess interval's CDF was modeled as a linear combination of the CDFs of the four river grain-age distributions. For interval i ($i = 1, \dots, 4$) and at grid point j ($j = 1, \dots, 100$) the loess CDF value y_{ij} is represented by a random variable with mean

$$1x1j + 2ix2j + 3ix3j + 4ix4j$$

where x_{kj} represents the CDF value of river k ($k = 1, \dots, 4$) at grid point j . The coefficients $1i$, $2i$, $3i$, and $4i$ are the mixing weights for interval i and are constrained by the requirement that $kki = 1$ for each interval i .

The loess CDF random variable y_{ij} is assumed to be equal to the mean given above plus an error e_{ij} . Such an error represents a deviation of the loess CDF from the weighted sum of the river CDFs. Owing to the non-decreasing and right-continuous nature of CDFs, sequential errors are likely to be correlated. This autocorrelation is modeled using a first-order autoregressive process, i.e.

$$e_{ij} = \phi e_{i,j-1} + f_{ij}$$

where the innovations f_{ij} are assumed to be independent and normally distributed with mean 0 and constant variance 2. Higher orders of this autoregressive process were also considered, but did not improve upon the fit of the first-order process. Parameters and were assumed to remain constant across the four intervals and were estimated in a model that allowed all intervals to

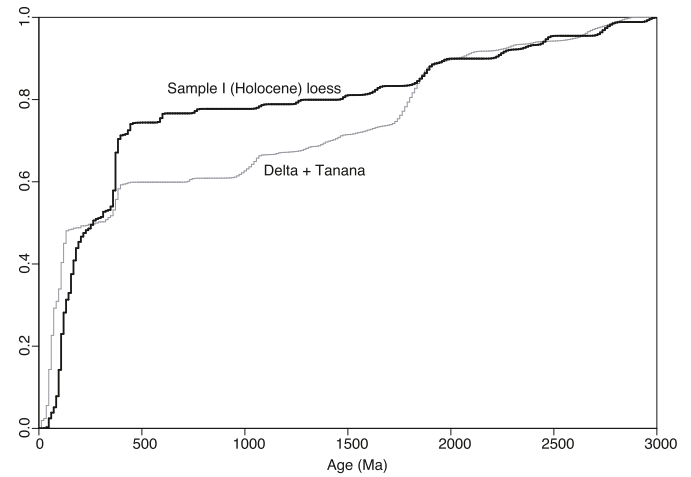


Fig. 5. One example of the cumulative distribution function (CDF) of detrital-zircon U–Pb ages from the Holocene loess interval plotted alongside the CDF of combined detrital-zircon U–Pb ages from two river samples (Delta and Tanana). No simple combination of river zircon grain ages matched the grain-age distribution from any loess interval.

simultaneously contribute information to the posterior distributions of these parameters.

The Bayesian model was fit in JAGS (Plummer, 2003) and incorporated the following uninformative prior distributions. The

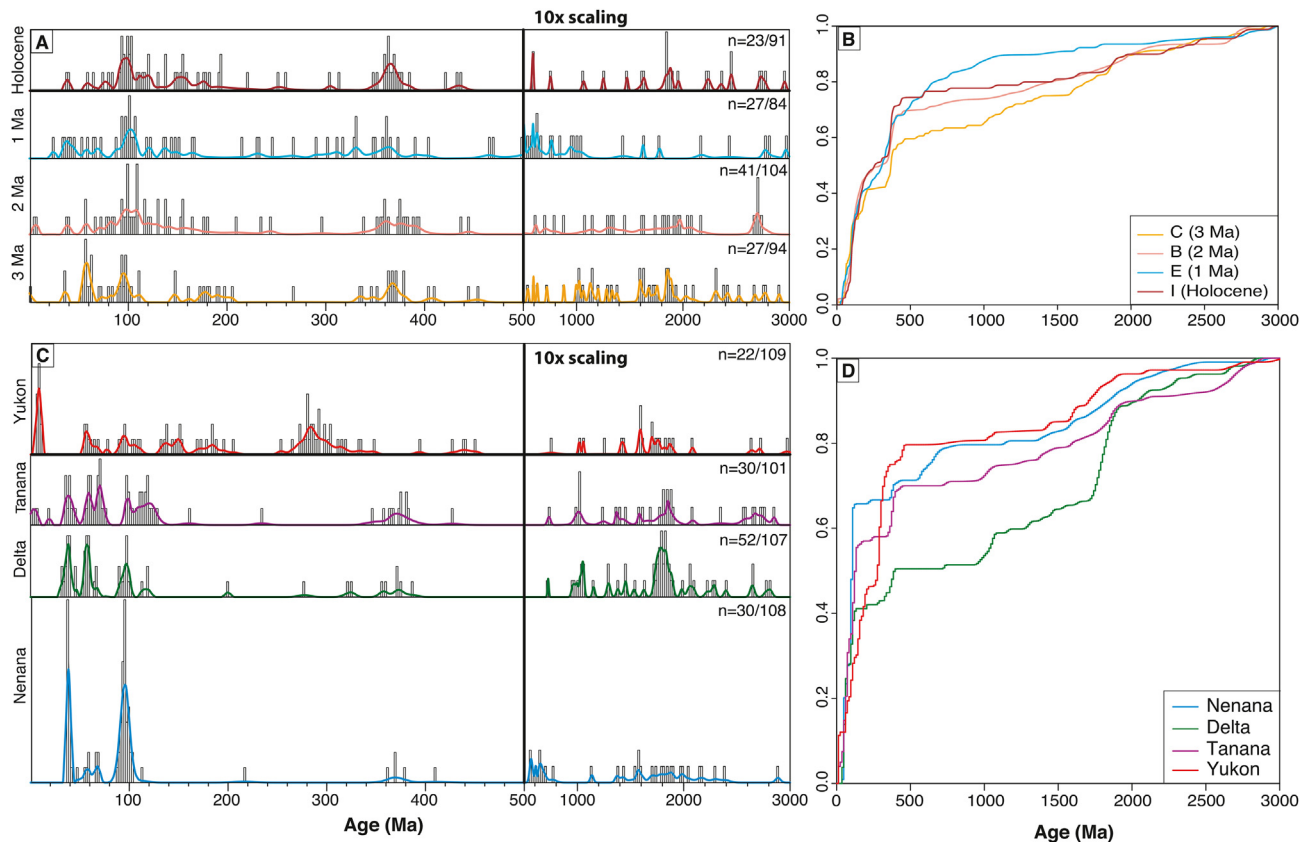


Fig. 4. Probability density plots (PDPs) and corresponding smoothed cumulative distribution functions (CDFs) of the detrital-zircon ages from the four Gold Hill loess samples (A, B) and the four rivers (C, D). PDPs were computed at each point of a regular grid of length 100 stretched across the span of 0–3000 Ma. Grain ages >500 Ma are scaled 10x to better show their peaks. Histograms plotted under the PDPs display grain counts on the density scale (i.e., the bar's cumulative area equals one). The shortest bars in each plot represent a single grain, and each spans 2 Myrs on the left-hand side, and 18 Myrs on the right-hand side. Smoothed CDFs were generated from the PDPs via numerical integration. See Fig. 1 for river sample locations. See also Table 3 for Anderson–Darling statistics of age similarities amongst all loess and all river sediment samples.

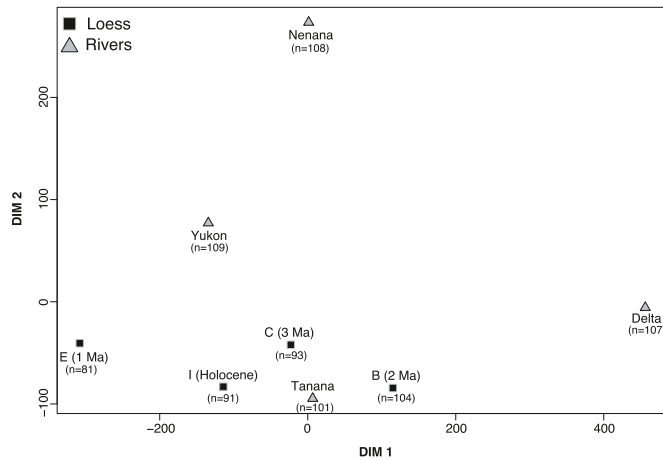


Fig. 6. Multi-dimensional scaling (MDS) plot to demonstrate how similar/dissimilar the detrital-zircon grain ages of all samples are. Loess intervals and river samples are labeled. In parentheses are the number of grain ages per sample. Plot constructed with the methodology applied to detrital-zircon chronology by Vermeesch (2013).

vector (1i 2i 3i 4i)T was jointly assumed to follow a Dirichlet distribution with hyperparameter vector (of length 4), which does not depend on i . Nuisance parameters and 2 were given uniform $(-1, 1)$ and half-Cauchy distributions, respectively. Each element in the vector was independently given a half-Cauchy distribution.

JAGS fits models using a Gibbs sampler (Geman and Geman, 1984) that may contain Metropolis-Hastings steps or slice sampling steps within. The resulting output is a long series of correlated draws from the posterior distributions of the model parameters. For this model we ran three separate chains of the Gibbs sampler at 30,000 iterations following a burn-in (to eliminate chains' dependence on their initial values) of 21,000 iterations. Trace plots and Gelman-Rubin convergence diagnostics (Gelman and Rubin, 1992) were assessed to verify convergence of the stochastic algorithms.

The results reported for the Bayesian linear models summarize parameters' posterior distributions by their mean and standard deviation.

3. Results

Both loess and river samples yielded grains that range in age from one million up to three billion years old (Supplementary Data Table 1). Fig. 4 shows PDPs and CDFs of all grain ages. The PDPs were generated by our custom PDP-generating code and integrated into the Bayesian model. Our ultimate goal was to calculate the balance of grain ages contributed to each loess interval from each river and determine if those contributions changed over time.

3.1. Gold Hill loess detrital zircon ages

Fig. 4a illustrates the significant peaks in detrital-zircon U–Pb ages of the loess samples, most of which are in the range of 30–400 Ma. Minor peaks are seen at ages >500 Ma, which are amplified ten-fold in order to better show sample similarities among the older grain ages.

All loess sample detrital-zircon ages share significant peaks ca. 40, 60, and 100. The amplitudes of these peaks vary; for example, sample C has a distinctly higher ca. 60 Ma peak. Other peaks in each sample do not overlap, for example grain populations at 370 and 1900 Ma are seen in only samples C and I. Sample differences may be more evident in the CDFs (Fig. 4b)—sample E differs the most, while samples C, B, and I correlate better in the 500–2000 Ma

range. The base and top of the Gold Hill loess deposit are the most similar. This is confirmed by the p -values of the Anderson-Darling test (Table 3), which are above 0.05 for all sample pairs, and is 0.81 for samples C and I. Sample E is the most distinct from sample I, suggesting that these two samples should show the greatest difference in sources for their detrital zircons. There are no grain-age peaks in any of the loess samples that are absent from the river samples.

3.2. Fairbanks-area river zircon ages

In contrast to the loess intervals, the four rivers we sampled have lower commonality in detrital-zircon ages (Fig. 4c and d), with peaks shared amongst all samples only at ca. 60 Ma and 90 Ma, and significant differences in the amplitudes of those peaks. The Delta, Tanana, and Nenana rivers additionally share a strong detrital-zircon age peak of ca. 40 Ma. The Tanana River has unique grain-age peaks at ca. 70 Ma and 120 Ma, as well as a <10 Ma population. The Yukon River appears the most different in the PDP (Fig. 4c; Table 3), with significant grain populations of ca. 10, 150, and 280 Ma. The CDF (Fig. 4d) indicates that the Yukon River grains are the most distinct from the other river grains at ages <500 Ma. The unique Yukon River 280 Ma age peak is not represented in any of the loess samples.

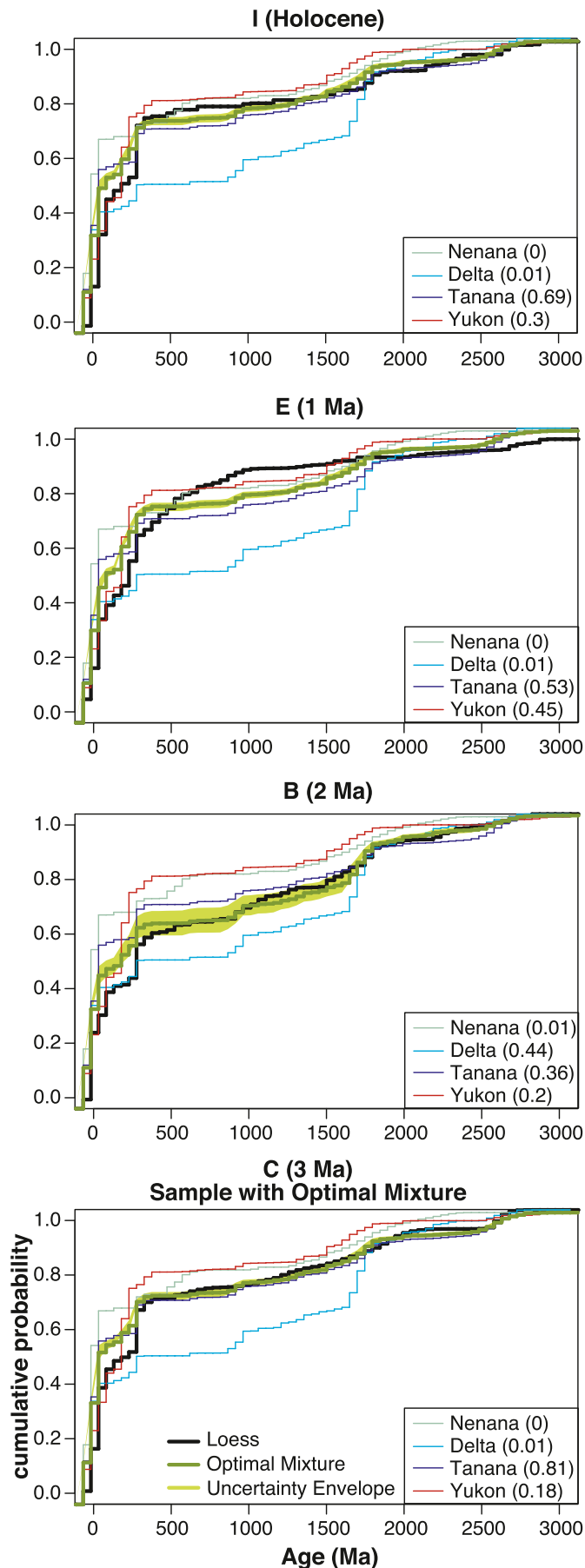
The Anderson-Darling test (Table 3) indicates that the zircon age populations of all the rivers except for the Delta and Tanana are statistically distinguishable at the 95% confidence interval. The Yukon has no commonality with the Delta and Nenana rivers (p -value = 0.00), and the Tanana vs. Yukon and Tanana vs. Nenana detrital-zircon sources have a very low probability of commonality (p -value = 0.03). Cumulatively, the Delta and Tanana rivers are the most similar, with a p -value of 0.08, as seen particularly in the abundance of grain ages younger than ca. 100 Ma (Fig. 4). This is not surprising, given that the Delta River is a tributary of the Tanana (Fig. 1).

3.3. River sources of Gold Hill loess

The Bayesian linear models described above yielded estimates of each river's contribution to the complete grain-age distributions of each loess interval. These are estimated by the MDS plot (Fig. 6), which is a 2D representation of sample similarities. The results are shown in more detail in the CDFs of Fig. 7 and reported along with 95% confidence intervals in Table 4. We cannot exclude the possibility that grains arrived from other sources; this model is a test of whether or not the loess sample zircon ages can be reconstructed by varying proportions of river zircons.

In our model, the Tanana River accounts for the majority of zircon grains in all loess intervals, with the Yukon River as the secondary major source. Nenana River sediment was absent to insignificant (0–8%, including the confidence interval) throughout the sampled loess intervals. The Delta River contribution is computed to be similarly absent to low (0–8%, including the confidence interval) in three of the four sampled loess intervals, which likely reflects its overlap with Tanana River sediments.

Sample C (ca. 3 Ma) shows a strong contribution of Tanana River grains ($82 \pm 12\%$, with a minimal ($17 \pm 11\%$) portion coming from the Yukon River. At 2 Ma (sample B), the model computes the only Delta River contribution of any sample, although the errors are much higher ($44 \pm 19\%$). If we combine the Tanana and Delta river sources, together these account for ~82% of zircons, and the Yukon River contribution was $16 \pm 12\%$. By 1 Ma (sample E), the Tanana River sediment contribution fell to $55 \pm 12\%$, and the Yukon River zircon portion rose significantly to $43 \pm 12\%$. Finally, in Holocene time (sample I), the Tanana River contribution rose to $71 \pm 12\%$, and



the Yukon River portion fell to $28 \pm 12\%$.

The strong contribution of the Tanana River sediments to the loess samples is also shown by the MDS plot of Fig. 6. Samples C, B, and I (3 Ma, 2 Ma, and Holocene) are seen as the closest relative to the Tanana River, while sample E (1 Ma) is a bit more distantly related and situated closer to the Yukon River zircon age population than the other samples. The Delta and Nenana rivers are not distinguished as significant contributors.

4. Discussion

A growing body of literature on loess deposition indicates that both fluvial and dust-storm processes are commonly involved (Aleinikoff et al., 1999; Újvári et al., 2012; Stevens et al., 2013; Licht et al., 2016; Shang et al., 2018; Sun et al., 2018; Nawrocki et al., 2018; Wang, 2019; He et al., 2020; Wu et al., 2021). Although our study cannot eliminate the potential for windborne sediment from any source, including reworked loess itself, the statistical approach described here presents a method that quantifies contributions from local river deposits during the 3 Myrs of accumulation of Gold Hill loess.

This approach estimated a mixture weight for the Yukon River of between 0.16 and 0.43 between the loess samples. The PDP in Fig. 4c indicates that the Yukon grain ages had a notable peak ca. 280 Ma, with no obvious corresponding peak in the loess samples. Similarly, the 10 Ma peak in the Yukon River and the <10 Ma peak in the Tanana River samples are missing from all loess intervals. These discrepancies may be due to a variety of causes. Firstly, there is no expectation that zircon grains from all sources within the Fairbanks area would make its way into the Gold Hill site. Secondly, it may signal a weakness in the statistical approach used to estimate the mixture weights. However, it is important to note that visual comparison is vulnerable to insufficiently accounting for geochronological errors. Future work should attempt to quantify the degree to which discrepancies between the optimal mixture CDFs and the loess CDFs can be attributed to potential causes of error, including sampling variability, age-measurement error, and the omission of an important source of grains.

4.1. River grain ages and histories

The difference in ages amongst the sampled rivers reflects the bedrock through which the rivers run. The Yukon River is most notably different from the others, with headwaters in significantly different bedrock. The rivers that drain the Alaska Range (Fig. 1) are more similar, and they ultimately drain into the Tanana River near the Gold Hill loess site. We sampled the Delta, Nenana, and Tanana rivers because their catchments are closest to the Gold Hill site. Although the Tanana and Delta rivers have different headwaters, the Delta River is a tributary to the Tanana east of Fairbanks (Fig. 1). Because the Anderson-Darling p -value for river zircon ages is 0.08 for the Delta and Tanana rivers, above but close to the p -value cutoff of 0.05 for low probability of commonality, we treated the Delta and Tanana river contributions to the loess as separate in the modeling. However, their proximity suggests that their contributions to the loess can be considered together.

A full analysis of the bedrock parent units for the detrital-zircon grains in Interior Alaska rivers is beyond the scope of this paper, but

Fig. 7. Cumulative distribution functions (CDFs) of detrital-zircon grain ages from each of the four loess samples (heavy black line), each of the four sampled rivers (brightly colored thin lines), and the optimal mixture of river samples that yields the closest match to the loess interval age distribution (heavy green line). The model result includes a 95% credible envelope in light green. The numbers in parentheses in the legend boxes show the percentage contribution of each river to the mixture.

Table 4

Posterior mean estimates of river mixture weights (with 95% credible intervals) for the four intervals. Mixture weights do not sum to 1 within intervals due to rounding errors.

Horizon C, ca. 3 Ma:			
Sample names	Relative contribution	95% CI lower bound	95% CI upper bound
Nenana age	0.00	0.00	0.02
Delta age	0.00	0.00	0.06
Tanana age	0.82	0.70	0.94
Yukon age	0.17	0.05	0.28
Horizon B, ca. 2 Ma:			
Sample names	Relative contribution	95% CI lower bound	95% CI upper bound
Nenana age	0.00	0.00	0.08
Delta age	0.45	0.26	0.64
Tanana age	0.37	0.18	0.57
Yukon age	0.16	0.03	0.28
Horizon E, ca. 1 Ma:			
Sample names	Relative contribution	95% CI lower bound	95% CI upper bound
Nenana age	0.00	0.00	0.02
Delta age	0.00	0.00	0.08
Tanana age	0.55	0.43	0.67
Yukon age	0.43	0.31	0.55
Horizon I, Holocene:			
Sample names	Relative contribution	95% CI lower bound	95% CI upper bound
Nenana age	0.00	0.00	0.02
Delta age	0.00	0.00	0.07
Tanana age	0.71	0.59	0.83
Yukon age	0.28	0.16	0.39

*Gibbs sampler details: Bayesian linear models fit with random initial values in 3 chains of 30,000 iterations (preceded by a burn-in of 20,000 iterations and 1000 additional adaptation iterations).

our new ages are consistent with those determined for detrital mica by Benowitz et al. (2019) for the same rivers investigated here. Notably, ca. 40 Ma grains make a significant age peak in the Nenana, Delta, and Tanana river samples (Fig. 4c), and subdued peaks in all loess samples (Fig. 4a), but are absent from the Yukon River sample. Headwaters of both the Nenana and Delta rivers originate within plutonic bedrock dated to ca. 40 Ma (Wilson et al., 2015). Our U–Pb zircon ages support detrital-mica age studies that indicate strong overlap between the Delta and Upper and Lower Tanana river sediment sources (Benowitz et al., 2019). This is not surprising, given that the Delta River flows into the Tanana River north of the Alaska Range and east of Fairbanks (Fig. 1). In contrast with the Nenana River, whose detrital-zircon grain ages are almost exclusively less than 100 Myrs, the Tanana and Delta rivers have a >120 Ma zircon grain-age peak (Fig. 4c) that may indicate a contribution from a Yukon–Tanana Uplands source (Fig. 1; Benowitz et al., 2019). The Yukon River sample ca. 280 Ma peak may reflect input from Pennsylvanian to Permian igneous bedrock that was unique to the Yukon–Tanana terrane (summarized in Table 2 of Piercey et al., 2006; Nelson et al., 2006).

The Nenana, Delta, and Tanana rivers are linked geographically and chronologically. Recent dating of detrital muscovite grains from the Nenana River indicates that the river has been flowing northward from headwaters in the Alaska Range since early Miocene time (Benowitz et al., 2019), thus carrying sediments locally for the duration of loess deposition. Previous studies suggested that the Delta River headwaters region of the eastern Alaska Range is younger (e.g., Begét and Keskinen, 2003), but recent studies support a model in which the Delta River reversed course and began flowing northward during mid-Miocene Alaska Range uplift (Benowitz et al., 2011; Finzel et al., 2015). Tanana River drainage history is likely linked with the Nenana and Delta river paleo-drainages. Additionally, the Wrangell volcanic field and Alaska Range have been significant topographic barriers since at

least 6 Ma (Benowitz et al., 2014, 2019; Brueseke et al., 2019; Berkelhammer et al., 2019). We therefore infer that the Tanana River has had similar bedrock sources over the past 3 Myrs, and confidently suggest that the Nenana, Delta, and Tanana rivers sampled in this study have been carrying sediment for the span of the loess intervals sampled.

The Yukon River has a more complicated paleodrainage history—there are, in effect, two “Yukon” rivers. The ancestral Yukon River occupied the Kwik’pak River Basin and was a large southwestward-draining system since the Pliocene (Ager et al., 1994; Vanlaningham et al., 2009). The Canadian “Yukon River” drained southward into the Gulf of Alaska until a glacially driven drainage reorganization and near-doubling of the drainage area related to the Cordilleran ice sheet maximum ca. 2.6 Ma (Duk-Rodkin et al., 2001, 2004; Hidy et al., 2013; Bender et al., 2019). The Yukon River was likely not carrying its modern load of sediments locally by the time of onset of deposition of the lower Gold Hill loess, and our oldest loess sample C has the lowest modern Yukon–River contribution. Rapid, deep incision of the Yukon–Tanana uplands by the Fortymile River tributary to the Yukon River tripled the Yukon sediment load to the Bering Sea from ca. 2.4–1.8 Ma (Bender et al., 2019), an interval that overlaps with our sample B. The small (20%) contribution of Yukon sediment to the ca. 2 Ma loess interval could be because (1) the Yukon River pre-2 Ma zircon “fingerprint” was different, (2) local wind redistribution lagged the arrival of the sediment pulse, (3) this was an interglacial period when the Yukon River floodplain was vegetated, or (4) winds from the north/northeast were less prevalent in this time interval. Interestingly, a river incision hiatus from 1.8 to 1 Ma (Horikawa et al., 2015; Bender et al., 2019) is not evident in 1 Ma loess sample E, which has the highest contribution (45%) of Yukon River grains. Or, perhaps sample E captures the ca. 1 Ma reinvigorated incision proposed by Bender et al. (2019) to reflect a major transition in Pleistocene climate cycles.

While our loess grain-age studies are relevant to estimating changes in local river sediment abundances over coarse intervals, finer-spaced sampling would help pinpoint transitions between river sources and perhaps even establish a lag time between sediment production and redistribution into the loess deposit.

4.2. Loess grain-age distributions and implications for paleo-wind directions

The heterogeneous age distributions of detrital-zircon grains in both the loess and the river deposits reflect broad sampling of rock formations in Alaska and Yukon, and make detrital-grain dating a robust approach to resolving mixed sources. We follow previous work that suggests the Yukon and Tanana rivers were the primary source of Late Quaternary loess in Interior Alaska (Muhs and Budahn, 2006). As we noted in the Introduction, there is no way to exclude contributions to the loess from wind-borne sediments from anywhere in Alaska. Still, the model we generated (Fig. 7), combined with previous studies (particularly Muhs and Budahn, 2006), support the hypothesis that U–Pb zircon ages from loess intervals in Interior Alaska can be reconstructed from U–Pb zircon ages from local river deposits.

In Figs. 7 and 8, we show the relative contributions from local river sediment sources through time (these values are also listed in Table 4). The ca. 3 Ma base of the Gold Hill loess deposit has an almost exclusively Tanana River sediment source, which indicates either deposition by primarily east–southeasterly winds or a very limited Yukon–River sediment source. If we combine the Delta–Tanana River contributions, this prevailing wind direction held until ca. 2 Ma. By ca. 1 Ma, a significant north–northeasterly component was added, with nearly half the sediment source to the loess coming from the Yukon River. The Holocene loess interval suggests a small drop in northeasterly winds, indicated by a reduction in the Yukon River component to ~25%. There is no indication of westerly wind, as indicated by a virtually 0% contribution of Nenana River sediments through the duration of loess deposition.

Our findings thus support those of Muhs and Budahn (2006), whose elemental analyses of late Quaternary portions of three loess sites near Fairbanks and silty alluvium from Tanana, Yukon, and Nenana riverbanks suggest a Tanana River source. However, Muhs and Budahn (2006) also concluded that heavy minerals were sourced from either the Nenana or Yukon rivers, and their multilinear regression analyses estimated river contributions of ~50% Tanana, ~30% Yukon, and ~17% Nenana. Our detrital-zircon grain-age distributions indicate that little to no Nenana–River contribution is necessary in the 3-Myr history of the Gold Hill loess.

Muhs and Budahn (2006) propose winter winds were dominantly from the north/northeast, and summer winds from the south/southeast. Our sample intervals, spaced by 1 Myr, are too

coarse to pick up seasonal changes in wind directions, but their different source contributions may be due to differences between glacial and interglacial conditions that affect wind strength and sediment supply. Our findings therefore support the idea that the bulk of loess transport occurs when sediment is uncovered by snow/ice, and therefore could take place during interglacials and/or in summer months, when modern measurements indicate stronger winds.

4.3. Comparison with modern winds

Muhs and Budahn (2006) computed dominant high-wind directions for January and July in the Fairbanks area from the northeast and from the south, respectively. They suggest winds change seasonally in Interior Alaska, and propose a modern dust-transporting wind profile of a wintertime pressure gradient pushing air from the northeast Canadian high to the Aleutian low, and weaker summertime air flow from the southwest driven by a strengthening Pacific subtropical high.

We computed the average monthly wind speeds and directions based on data collected from 1941 until today by five ASOS network stations (<https://mesonet.agron.iastate.edu/>) in the Fairbanks area. This recent record shows a remarkably steady average wind pattern, with the strongest local winds in May and June and winds throughout the year coming predominantly from the south, but slightly more from the east between September and March, and slightly more from the west from June to August (Fig. 9; Supplementary Fig. 1; Supplementary Data Table 2). This contrasts somewhat with the seasonal wind directions presented by Lagroix and Banerjee (2004a, 2004b) for 1968–1996 (Fig. 9b). The ASOS wind data from that interval indicates a much greater seasonal variation on the shorter timescale, with southeasterly summer winds and northeasterly winter winds (Fig. 9b). This discrepancy is not too surprising given the range shown in the larger data set—wind directions vary greatly over the timescale of nearly a century. The safest conclusion to draw is that there has been no significant northwesterly component of wind in Fairbanks since 1941.

Interior Alaska, in particular the region around Fairbanks, has a remarkably low annual average wind speed of 1.9 ± 0.7 m/s. Currently, the peak dust-transporting season in Interior Alaska is during the May Chinook (or föhn) winds (Figs. 2 and 9; Supp. Fig. 1). These winds were called “snow eater” by the Niitsitapi (Blackfoot) people, because they dump moisture on windward slopes, then warm adiabatically as they descend the leeward side. Published studies on transport velocities for loess-sized particles indicate 4–14 m/s is required (Péwé, 1975; Nickling, 1978). The average May windspeed in the Fairbanks region since 1941 is 2.7 ± 2 m/s—this is quite low, and the maximum daily windspeed record is 4.7 m/s. However, even a daily record can be remarkably misleading, as gusts from a particular June windstorm were as high as 14 m/s

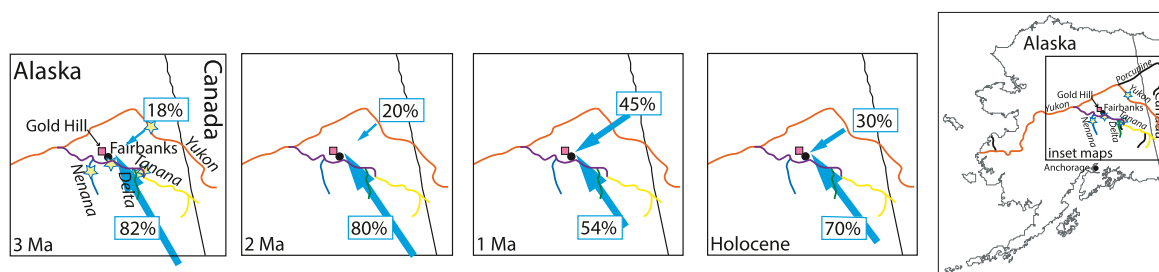


Fig. 8. Interpretations of river sources to Interior Alaska loess, based on the modeled contributions of detrital zircons from local river sample sites to the four dated loess intervals at the Gold Hill site (Fig. 7), at the intervals 3 Ma, 2 Ma, 1 Ma, and Holocene. The blue arrows are scaled to their relative contributions, with percentages shown.

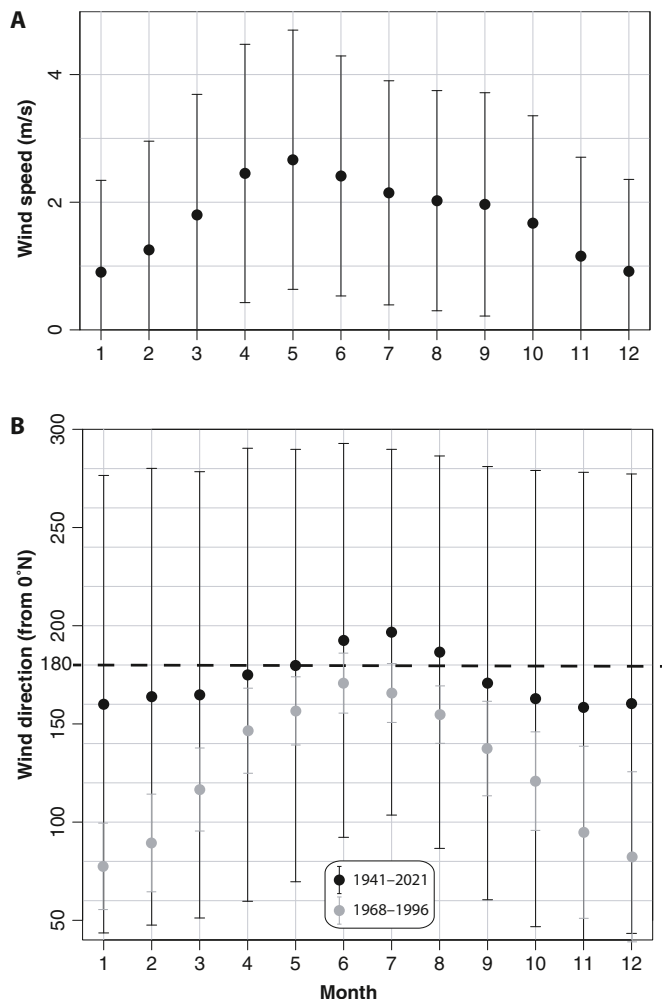


Fig. 9. Graphs showing the mean of the averaged wind speeds (top) and direction (bottom) for five weather stations near Fairbanks, AK. Black dots are data from February 1941 and November 2020, and grey dots are a subset of data from 1968 to 1996, chosen to overlap the study of Lagroix and Banerjee (2004b). Error bars are standard deviations. Data from <https://mesonet.agron.iastate.edu/>.

(Begét, 1996). This rapidly fluctuating wind landscape suggests that pinpointing periods of loess transport and deposition is not straightforward. However, our interpretation of loess contribution to Interior Alaska is consistent with the general patterns of wind direction over the past century. While the 3–2 Ma detrital-zircon record within the loess records dominantly southerly/southeasterly winds, northeasterly winds are certainly present at shorter time intervals and would have accommodated the different-sourced sediment input seen from 1 Ma onward.

Finer-spaced sampling, particularly associated with locally developed paleosol horizons, could aid in understanding the changing supply dynamics and place loess accumulation in the context of individual glacial–interglacial episodes.

5. Conclusions

Our new dataset of detrital-zircon ages in both Gold Hill loess intervals and river deposits of Interior Alaska improves our understanding of the sources of the loess. Using a binary mixture of riverbank deposits of the Tanana, Delta, Nenana, and Yukon rivers, we determined that a weighted combination of river sources fits

the zircon ages of the loess better than any single source (i.e., no single source matches age distributions in any loess interval). Based on spatial and temporal heterogeneities of zircon ages, we infer that the Tanana River was the main source of wind-transported sediment over the past 3 Myrs. The Yukon River contributed a significant portion of dust to the Gold Hill deposit after 1 Ma, possibly due to an interval of rejuvenated river incision, or prolonged northeasterly wind or other climatic conditions that favored dust transport from that source. Modern winds, averaged over many decades, are predominantly from the south/southeast, with strongest winds in May and June, however there are intervals of northeasterly winds. Although it's impossible to reconstruct seasonal wind patterns of the late Cenozoic from our loess grain-age data, our analyses suggest that prevailing winds over the past 3 Myrs are consistent with modern winds, and consistent with the main contributors to the vast loess deposits of Interior Alaska.

Author contributions

Elisabeth Nadin—secured funding, originated project, formulated original concept with J. Benowitz, formulated idea behind statistical method with S. Goddard, wrote paper; did all but the following. Scott Goddard—devised main statistical approach described in Methods section; generated Figs. 4–7, 9. Jeff Benowitz—brought Nadin to sampling location; assisted in sampling and sample processing; helped conceive approach of comparing river samples with loess samples; contributed to discussion on history of the Yukon River. Paul O'Sullivan—dated all detrital zircon grains; provided grain dating methodology (supplement). All authors have read and approved the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We gratefully acknowledge funding for this project provided by the Alaska Space Grant Program. Sampling in the field was assisted by Rebekah Tsigonis and Sam VanLaningham. Tsigonis also helped process samples. Detrital-zircon data and statistical analysis conversations with Alexander Tye and Joel Saylor helped us craft our approach and understand the limitations of our data sets. Anonymous reviews were instrumental in helping us craft this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.quascirev.2021.107266>.

References

- Ager, T.A., Matthews Jr., J.V., Yeend, W., 1994. Pliocene terrace gravels of the ancestral Yukon River near Circle, Alaska: palynology, paleobotany, paleo-environmental reconstruction and regional correlation. *Quat. Int.* 22, 185–206.
- Alder, J.R., Hostetler, S.W., 2015. Global climate simulations at 3000-year intervals for the last 21,000 years with the GENMOM coupled atmosphere–ocean model. *Clim. Past* 11, 449–471. <https://doi.org/10.5194/cp-11-449-2015>, 2015.
- Aleinikoff, J.N., Muhs, D.R., Sauer, R., Fanning, C.M., 1999. Late Quaternary loess in northeastern Colorado, Part II — Pb isotopic evidence for the variability of loess sources. *Geol. Soc. Am. Bull.* 111, 1876–1883.
- Aleinikoff, J.N., Muhs, D.R., Bettis III, E.A., Johnson, W.C., Fanning, C.M., Benton, R., 2008. Isotopic evidence for the diversity of late Quaternary loess in Nebraska: glaciogenic and nonglaciogenic sources. *Geol. Soc. Am. Bull.* 120, 1362–1377. <https://doi.org/10.1130/B26222.1>.
- Begét, J.E., 1988. Tephra and sedimentology of frozen Alaskan loess. In:

- Senneset, Kaare (Ed.), International Conference on Permafrost, 5, Proceedings, V. 1, Trondheim, Norway, pp. 672–677, Aug. 2–5, 1988.
- Begét, J., 1996. Tephrochronology and paleoclimatology of the last interglacial-glacial cycle recorded in Alaskan loess deposits. *Quat. Int.* 34–36, 121–126.
- Begét, J.E., 2001. Continuous late Quaternary proxy climate records from loess in Beringia. *Quat. Sci. Rev.* 20, 499–507.
- Begét, J., Hawkins, D., 1989. Influence of orbital parameters on Pleistocene loess deposition in central Alaska. *Nature* 337, 151–153.
- Begét, J.E., Keskinen, M.J., 2003. Trace-element geochemistry of individual glass shards of the Old Crow tephra and the age of the Delta glaciation, central Alaska. *Quat. Res.* 60, 63–69. [https://doi.org/10.1016/S0033-5894\(03\)00095-4](https://doi.org/10.1016/S0033-5894(03)00095-4).
- Begét, J.E., Stone, D.B., Hawkins, D.B., 1990. Paleoclimatic forcing of magnetic susceptibility variations in Alaskan loess during the late Quaternary. *Geology* 18, 40–43.
- Bender, A.M., Lease, R.O., Corbett, L.B., Bierman, P., Caffee, M.W., 2019. Ongoing bedrock incision of the Fortymile River driven by pliocene–pleistocene Yukon river capture, eastern Alaska, USA, and Yukon, Canada. *Geology* 46, 635–638. <https://doi.org/10.1130/G40203.1>.
- Benowitz, J.A., Layer, P.W., Armstrong, P., Perry, S.E., Haeussler, P.J., Fitzgerald, P.G., VanLaningham, S., 2011. Spatial variations in focused exhumation along a continental-scale strike-slip fault: the Denali fault of the eastern Alaska Range. *Geosphere* 7, 455–467.
- Benowitz, J.A., Layer, P.W., Vanlaningham, S., 2014. Persistent long-term (c. 24 Ma) exhumation in the Eastern Alaska Range constrained by stacked thermochronology. *Geol. Soc., Lond. Spec. Publ.* 378, 225–243.
- Benowitz, J.A., Davis, K., Roeske, S., 2019. A river runs through it both ways across time: $^{40}\text{Ar}/^{39}\text{Ar}$ detrital and bedrock muscovite geochronology constraints on the Neogene paleodrainage history of the Nenana River system. *Alaska Range: Geosphere* 15, 682–701. <https://doi.org/10.1130/GES01673.1>.
- Berger, G.W., 2003. Luminescence chronology of late Pleistocene loess-paleosol and tephra sequences near Fairbanks, Alaska. *Quat. Res.* 60, 70–83.
- Berger, G.W., Péwé, T.L., 2001. Last interglacial age of the Eva Forest Bed, central Alaska, from thermoluminescence dating of bracketing loess. *Quat. Sci. Rev.* 20, 485–498.
- Berkelhammer, S.E., Brueseke, M.E., Benowitz, J.A., Trop, J.M., Davis, K., Layer, P.W., Weber, M., 2019. Geochemical and geochronological records of tectonic changes along a flat-slab arc-transform junction: circa 30 Ma to ca. 19 Ma Sonya Creek volcanic field, Wrangell Arc, Alaska. *Geosphere* 15, 1508–1538.
- Brueseke, M.E., Benowitz, J.A., Trop, J.M., Davis, K.N., Berkelhammer, S.E., Layer, P.W., Morter, B.K., 2019. The Alaska Wrangell Arc: ~ 30 Ma of subduction-related magmatism along a still active arc-transform junction. *Terra. Nova* 31, 59–66.
- Che, X., Li, G., 2013. Binary sources of loess on the Chinese Loess Plateau revealed by U–Pb ages of zircon. *Quat. Res.* 80, 545–551.
- Collins, F.R., 1985. Map Showing a Vegetated Dune Field in Central Alaska. US Geological Survey, Miscellaneous Field Studies, Map. MF-1708, scale 1 : 250 000.
- Davies, L.J., Jensen, B.J.L., Froese, D.G., Wallace, K.L., 2016. Late Pleistocene and Holocene tephrostratigraphy of interior Alaska and Yukon: key beds and chronologies over the past 30,000 years. *Quat. Sci. Rev.* 146, 28–53. <https://doi.org/10.1016/j.quascirev.2016.05.026>.
- Dickinson, W.R., Gehrels, G.E., 2003. U–Pb ages of detrital zircons from Permian and Jurassic eolian sandstones of the Colorado Plateau, USA: paleogeographic implications. *Sediment. Geol.* 163, 29–66. [https://doi.org/10.1016/S0037-0738\(03\)00158-1](https://doi.org/10.1016/S0037-0738(03)00158-1).
- Donelick, R.A., O'Sullivan, P.B., Ketcham, R.A., 2005. Apatite Fission-Track Analysis. *Reviews in Mineralogy and Geochemistry*, vol. 58. Mineralogical Society of America, pp. 49–94.
- Dortch, J.M., Owen, L.A., Caffee, M.W., Li, D., Lowell, T.V., 2010. Beryllium-10 surface exposure dating of glacial successions in the Central Alaska Range. *J. Quat. Sci.* 25, 1259–1269. <https://doi.org/10.1002/jqs.1406>.
- Duk-Rodkin, A., Barendregt, R.W., White, J.M., Singhroy, V.H., 2001. Geologic evolution of the Yukon river: implications for placer gold. *Quat. Int.* 82, 5–31. [https://doi.org/10.1016/S1040-6182\(01\)00006-4](https://doi.org/10.1016/S1040-6182(01)00006-4).
- Evans, M.E., Jensen, B.J.L., Kravchinsky, V.A., Froese, D.G., 2011. The Kamikatsura event in the gold hill loess, Alaska. *Geophys. Res. Lett.* 38, L13302. <https://doi.org/10.1029/2011GL047793>.
- Froese, D.G., Smith, D.G., Clement, D.T., 2005. Characterizing large river history with shallow geophysics: middle Yukon River, Yukon Territory and Alaska. *Geomorphology* 67, 391–406.
- Froese, D., Zazula, G.D., Reyes, A.V., 2006. Seasonality of the late Pleistocene Dawson tephra and exceptional preservation of a buried vegetation surface in central Yukon Territory. *Quat. Sci. Rev.* 25, 1542–1551.
- Gehrels, G., 2012. Detrital zircon U–Pb geochronology: current methods and new opportunities. *Tectonics Sediment. Basins: Recent Adv.* 45–62.
- Geman, S., Geman, D., 1984. Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. *IEEE Trans. Pattern Anal. Mach. Intell.* 6, 721–741.
- Gelman, A., Rubin, D.B., 1992. Inference from iterative simulation using multiple sequences. *Stat. Sci.* 7, 457–472.
- Gong, H., Xie, W., Zhang, R., Zhang, Y., 2017. U–Pb ages of detrital zircon and provenances of Red Clay in the Chinese Loess Plateau. *J. Asian Earth Sci.* 138, 495–501. <https://doi.org/10.1016/j.jseas.2017.02.014>.
- Guthrie, R.D., 1990. Frozen Fauna of the Mammoth Steppe: the Story of Blue Babe. University of Chicago Press, Chicago, Ill.
- He, Y., Li, G., Huang, X., Wang, X., Zhang, Y., 2020. Loess depositional dynamics and paleoclimatic changes in the Yili Basin, Central Asia, over the past 250 ka. *Catena* 195. <https://doi.org/10.1016/j.catena.2020.104881>.
- Hidy, A.J., Gosse, J.C., Froese, D.G., Bond, J.D., Rood, D.H., 2013. A latest Pliocene age for the earliest and most extensive Cordilleran Ice Sheet in northwestern Canada. *Quat. Sci. Rev.* 61, 77–84.
- Hopkins, D.M., Matthews, J.V., Schweger, C.D., Young, S.B. (Eds.), 1982. *Paleoecology of Beringia*. Academic Press, New York.
- Jensen, B.J.L., Evans, M.E., Froese, D.G., Kravchinsky, V.A., 2016. 150,000 years of loess accumulation in central Alaska. *Quat. Sci. Rev.* 135, 1–23. <https://doi.org/10.1016/j.quascirev.2016.01.001>.
- Lagroix, F., Banerjee, S.K., 2002. Paleowind directions from the magnetic fabric of loess profiles in central Alaska. *Earth Planet Sci. Lett.* 195, 99–112.
- Lagroix, F., Banerjee, S.K., 2004a. Cryptic post-depositional reworking in aeolian sediments revealed by the anisotropy of magnetic susceptibility. *Earth Planet Sci. Lett.* 224, 453–459.
- Lagroix, F., Banerjee, S.K., 2004b. The regional and temporal significance of primarily aeolian magnetic fabrics preserved in Alaskan loess. *Earth Planet Sci. Lett.* 225, 379–395.
- Lea, P.D., 1996. Vertebrate tracks in Pleistocene eolian sand-sheet deposits of Alaska. *Quat. Res.* 45, 226–240.
- Lea, P.D., Waythomas, C.F., 1990. Late-Pleistocene eolian sand sheets in Alaska. *Quat. Res.* 34, 269–281.
- Licht, A., Pullen, A., Kapp, P., Abell, J., Giesler, N., 2016. Eolian cannibalism: reworked loess and fluvial sediment as the main sources of the Chinese Loess Plateau. *Geol. Soc. Am. Bull.* 128, 944–956. <https://doi.org/10.1130/B31375.1>.
- Liu, W., Liu, Z., An, Z., Sun, J., Chang, H., Wang, N., Dong, J., Wang, H., 2014. Late Miocene episodic lakes in the arid Tarim Basin, western China. *Proc. Natl. Acad. Sci. Unit. States Am.* 111, 16292–16296.
- Muhs, D.R., Budahn, J.R., 2006. Geochemical evidence for the origin of late Quaternary loess in central Alaska. *Can. J. Earth Sci.* 43, 323–337. <https://doi.org/10.1139/E05-115>.
- Muhs, D.R., Ager, T.A., Bettis III, E.A., McGeehin, J., Been, J.M., Begét, J.E., Pavich, M.J., Stafford Jr., T.W., Pinney, D., 2003. Stratigraphy and paleoclimatic significance of late Quaternary loess-paleosol sequences of the last interglacial-glacial cycle in central Alaska. *Quat. Sci. Rev.* 22, 1947–1986.
- Muhs, D.R., Pigati, J.S., Budahn, J.R., Skipp, G.I., Bettis III, A., Jensen, B., 2018. Origin of last-glacial loess in the western Yukon–Tanana Upland, central Alaska, USA. *Quat. Res.* 89, 1–23. <https://doi.org/10.1017/qua.2018.11>.
- Nawrocki, J., Gozhik, P., Łanczont, M., Pańczyk, M., Komar, M., Bogucki, A., Williams, I.S., Czupry, Z., 2018. Palaeowind directions and sources of detrital material archived in the Roxolany loess section (southern Ukraine). *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 496, 121–135.
- Nelson, J.L., Colpron, M., Piercey, S.J., Dusel-Bacon, C., Murphy, D.C., Roots, C.F., 2006. Paleozoic tectonic and metallogenic evolution of the pericratonic terranes in Yukon, northern British Columbia and eastern Alaska. In: Colpron, M., Nelson, J.L. (Eds.), *Paleozoic Evolution and Metallogeny of Pericratonic Terranes at the Ancient Pacific C Margin of North America, Canadian and Alaskan Cordillera: Geological Association of Canada Special Paper*, vol. 45, pp. 323–360.
- Péwé, T.L., 1955. Origin of the upland silt near Fairbanks, Alaska. *Geol. Soc. Am. Bull.* 66, 699–724.
- Péwé, T.L., 1975. *Quaternary Geology of Alaska*. US Geological Survey, Professional Paper 835.
- Péwé, T., Westgate, J., Preece, S., Brown, P., Leavitt, S.W., 2009. Late Pliocene Dawson cut forest bed and new tephrochronological findings in the gold hill loess, east-central Alaska. *Geol. Soc. Am. Bull.* 121, 294–320.
- Piercey, S.J., Nelson, J.L., Colpron, M., Dusel-Bacon, C., Roots, C.F., Simard, R.-L., 2006. Paleozoic magmatism and crustal recycling along the ancient Pacific margin of North America, northern Cordillera. In: Colpron, M., Nelson, J.L. (Eds.), *Paleozoic Evolution and Metallogeny of Pericratonic Terranes at the Ancient Pacific Margin of North America, Canadian and Alaskan Cordillera: Geological Association of Canada Special Paper*, vol. 45, pp. 281–322.
- Plummer, M., 2003. JAGS: a program for analysis of Bayesian graphical models using Gibbs sampling. *Proc. 3rd Int. Workshop Distrib. Statist. Comput.* 124, 1–10.
- Preece, S., Westgate, J., Stemper, B., Pewe, T., 1999. Tephrochronology of late Cenozoic loess at Fairbanks, central Alaska. *Geol. Soc. Am. Bull.* 111, 71–90.
- Preece, S.J., Pearce, N.J.G., Westgate, J.A., Froese, D.G., Jensen, B.J.L., Perkins, W.T., 2011. Old Crow tephra across eastern Beringia: a single cataclysmic eruption at the close of Marine Isotope Stage 6. *Quat. Sci. Rev.* 30, 2069–2090.
- Pullen, A., Kapp, P., McCallister, A.T., Chang, H., Gehrels, G.E., Garzone, C.N., Heermance, R.V., Ding, L., 2011. Qaidam Basin and northern Tibetan Plateau as dust sources for the Chinese Loess Plateau and paleoclimatic implications. *Geology* 39, 1031–1034.
- Saylor, J.E., Sundell, K.E., 2016. Quantifying comparison of large detrital geochronology data sets. *Geosphere* 12, 203–220. <https://doi.org/10.1130/GES01237.1>.
- Saylor, J.E., Sundell, K.E., Sharman, G.R., 2019. Characterizing sediment sources by non-negative matrix factorization of detrital geochronological data. *Earth Planet Sci. Lett.* 512, 46–58.
- Shang, Y., Prins, M.A., Beets, C.J., Kaakinen, A., Lahaye, Y., Dijkstra, N., Rits, D.S., Wang, B., Zheng, H., van Balen, R.T., 2018. Aeolian dust supply from the Yellow River floodplain to the Pleistocene loess deposits of the Mangshan Plateau, central China: evidence from zircon U–Pb age spectra. *Quat. Sci. Rev.* 182, 131–143.
- Soreghan, M.J., Soreghan, G.S., Hamilton, M.A., 2012. Paleowinds inferred from detrital-zircon geochronology of upper Paleozoic loessite, western equatorial Pangea. *Geology* 30, 695–698.

- Stevens, T., Palk, C., Carter, A., Lu, H., Clift, P.D., 2010. Assessing the provenance of loess and desert sediments in northern China using U–Pb dating and morphology of detrital zircons. *Geol. Soc. Am. Bull.* 122, 1331–1344.
- Stevens, T., Carter, A., Watson, T.P., Vermeesch, P., Andò, S., Bird, A.F., Lu, H., Garzanti, E., Cottam, M.A., Sevastjanova, I., 2013. Genetic linkage between the Yellow River, the mu us desert and the Chinese Loess Plateau. *Quat. Sci. Rev.* 78, 355–368. <https://doi.org/10.1016/j.quascirev.2012.11.032>.
- Sun, J., Ding, Z., Xia, X., Sun, M., Windley, B.F., 2018. Detrital zircon evidence for the ternary sources of the Chinese Loess Plateau. *J. Asian Earth Sci.* 155, 21–34. <https://doi.org/10.1016/j.jseas.2017.10.012>.
- Sundell, K.E., Saylor, J.E., 2017. Unmixing detrital geochronology age distributions. *G-cubed* 18, 2872–2886.
- Thorson, R.M., Bender, G., 1985. Eolian deflation by ancient katabatic winds: a late Quaternary example from the north Alaska Range. *Geol. Soc. Am. Bull.* 96, 702–709.
- Tye, A.R., Wolfe, A.S., Niemi, N.A., 2019. Bayesian population correlation: a probabilistic approach to inferring and comparing population distributions for detrital zircon ages. *Chem. Geol.* 518, 67–78.
- Újvári, G., Varga, A., Ramos, F.C., Kovács, J., Németh, T., Stevens, T., 2012. Evaluating the use of clay mineralogy, SrNd isotopes and zircon UePb ages in tracking dust provenance: an example from loess of the Carpathian Basin. *Chem. Geol.* 304–305, 83–96.
- Vanlaningham, S., Pisias, N.G., Duncan, R.A., Clift, P.D., 2009. Glacial–interglacial sediment transport to the meiji drift, northwest pacific ocean: evidence for the timing of beringian outwashing. *Earth Planet. Sci. Lett.* 277, 64–72. <https://doi.org/10.1016/j.epsl.2008.09.033>.
- Vermeesch, P., 2004. How many grains are needed for a provenance study? *Earth Planet. Sci. Lett.* 224, 441–451. <https://doi.org/10.1016/j.epsl.2004.05.037>.
- Vermeesch, P., 2013. Multi-sample comparison of detrital age distributions. *Chem. Geol.* 341, 140–146. [10.1016/j.chemgeo.2013.01.010](https://doi.org/10.1016/j.chemgeo.2013.01.010).
- Vermeesch, P., 2018. Dissimilarity measures in detrital geochronology. *Earth Sci. Rev.* 178, 310–321.
- Wang, H., 2019. First preliminary evidence of Holocene loess in the Illinois River valley. *Quat. Sci. Rev.* 216, 124–126. <https://doi.org/10.1016/j.quascirev.2019.06.018>.
- Westgate, J., Stemper, B., Pewe, T., 1990. A 3 m.y. record of Pliocene–Pleistocene loess in interior Alaska. *Geology* 18, 858–861.
- Wilson, F.H., Hults, C.P., Mull, C.G., Karl, S.M., 2015. Geologic Map of Alaska, vol. 3340. U.S. Geological Survey Scientific Investigations Map. <https://doi.org/10.3133/sim3340> pamphlet 196 pp., 2 sheets, scale 1:1,584,000.
- Wu, C., Zheng, X., Zhou, L., Ren, S., Peng, Q., 2021. Quantitative estimation of provenance contributions to loess deposits in eastern China and implication for paleo-dust storm activity. *Geomorphology* 373. <https://doi.org/10.1016/j.geomorph.2020.107489>.
- Xie, J., Yang, S., Ding, Z., 2012. Methods and application of using detrital zircons to trace the provenance of loess. *Sci. China Earth Sci.* 55, 1837–1846.